روش‌های پیشرفته نمونه برداري در جنگل

نمونه برداري نقطه‌اي
Point sampling

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Two approaches for selecting individual into the sampling unite

1. **Probability proportional to frequency**

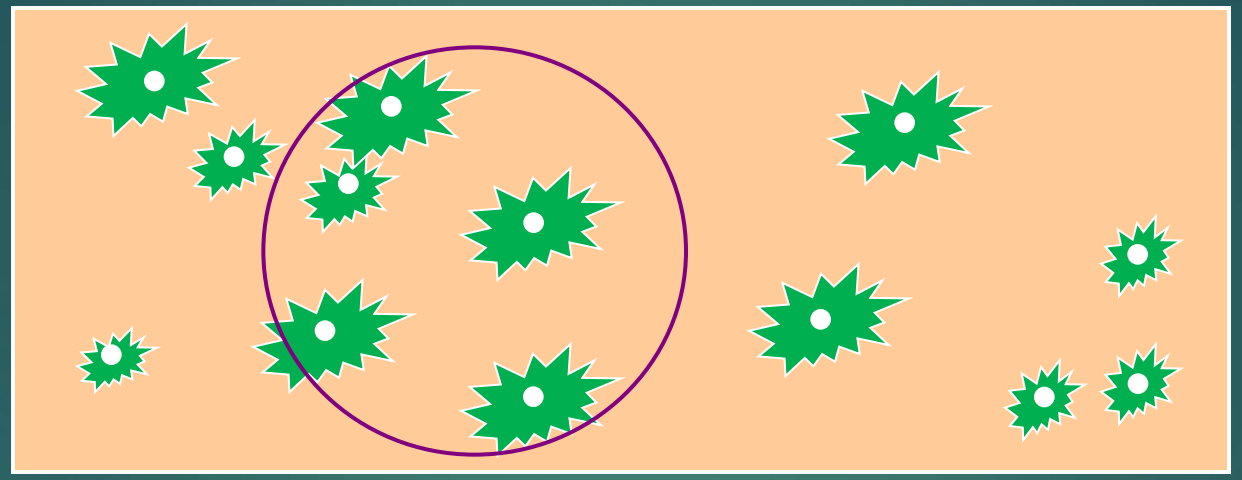
Vs.

2. **Probability proportional to size**

Comparing the approaches

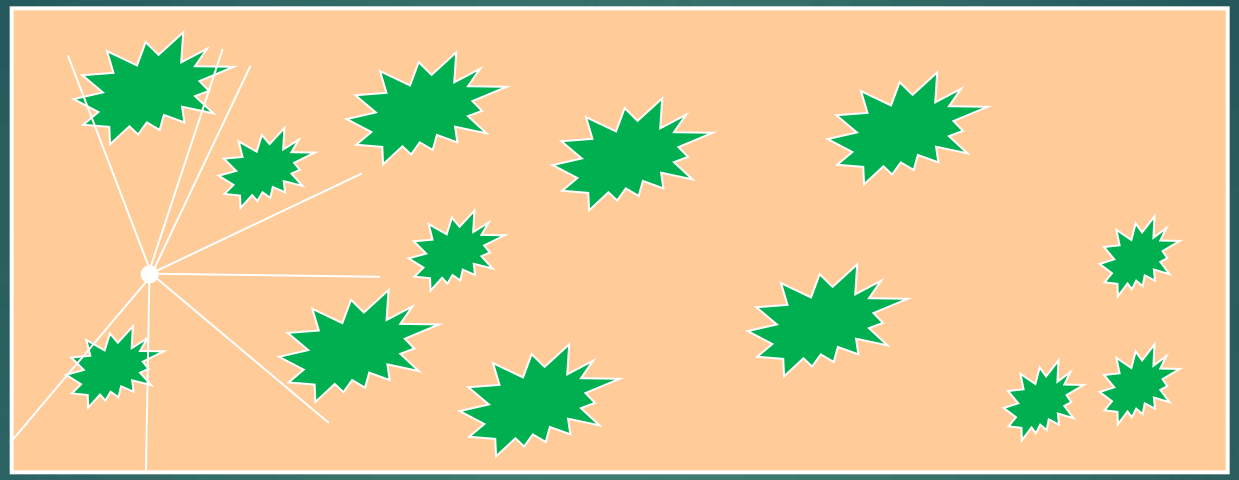
	PPF	PPF
Sampling is based on..	Frequency of occurrence	Tree size
Expansion factor	Same for all trees	Varies by tree

Sampling proportional to frequency



Fixed Area Plots – each tree has an equal chance of being selected regardless of its size. Each tree represents the same number of trees/ha.

Sampling proportional to size



Point Sampling – probability of a given tree being sampled is proportional to its size. Each selected tree represents a [potentially] different number of trees/ha.

Methods for Sampling Proportional to [Size]

1. List Sampling
2. PPS Sampling
3. Probability Proportional to Prediction

List Sampling

Listed sampling frame as used for „list sampling" where the selection probability is determined proportional to size.

Population element	List of the size variables of the population elements	List of cumulative sums	Assigned range
1	10	10	0 - 10
2	20	30	> 10 - 30
3	30	60	> 30 - 60
4	60	120	> 60 - 120
5	100	220	> 120 - 220

This sampling approach is also called list sampling because the selection can most easily be explained by listing the size variables and select from the cumulative sum with uniformly distributed random numbers (which perfectly simulates the unequal probability selection process). This is illustrated in Table 1: the size variables of the 5 elements are listed (not necessarily any order!) and the cumulative sums are calculated. The uniformly distributed random number is drawn between the lowest and highest possible value of that range, that is from 0 to the total sum. Assume, for example, the random number 111.11 is drawn; this falls into the range ">60 – 120" so that element 4 is selected. Obviously, the elements have then a selection probability proportional to the size variable.

مدل مفهومي نمونه برداري

- Protocol

- طراحي نمونه برداري
  Sampling design
  - روش نمونه برداري
    Sampling Method
    - نمونه برداري نقطه‌اي Point Sampling
    - نمونه برداري فاصله‌اي Distance Sampling
    - نمونه برداري خطي Line Sampling
    - نمونه برداري قطعه‌اي Plot Sampling
  - الگوي نمونه برداري
    Sampling patterns
    - تصادفي ساده Simple Random S.
    - بلوك تصادفي Block Random S.
    - سيستماتيک تصادفی Systematic S.
  - اجتماع نمونه
    Sampling clusering
    - نمونه واحد Single s.
    - نمونه خوشه‌اي Cluster S.
  - استراتژي نمونه برداري
    Sampling strategies
    - مونه بندي
      - پيشا مونه بندي(Prestratification)
      - پسا مونه بندي (Poststratification)
    - نمونه برداري چند گانه
      Multiphase S.
      - Two phase sampling with regression estimator
      - Two phase sampling with ratio estimator
      - Two phase sampling for stratification
      - .....
    - نمونه برداري چند گامي
      Multistage S.
      - Two stage sampling
      - Three stage sampling
      - .....

Point Sampling

Alternate names: angle count, variable plot cruising, plotless cruising

A method of sampling based on tree size, rather than its frequency of occurrence.

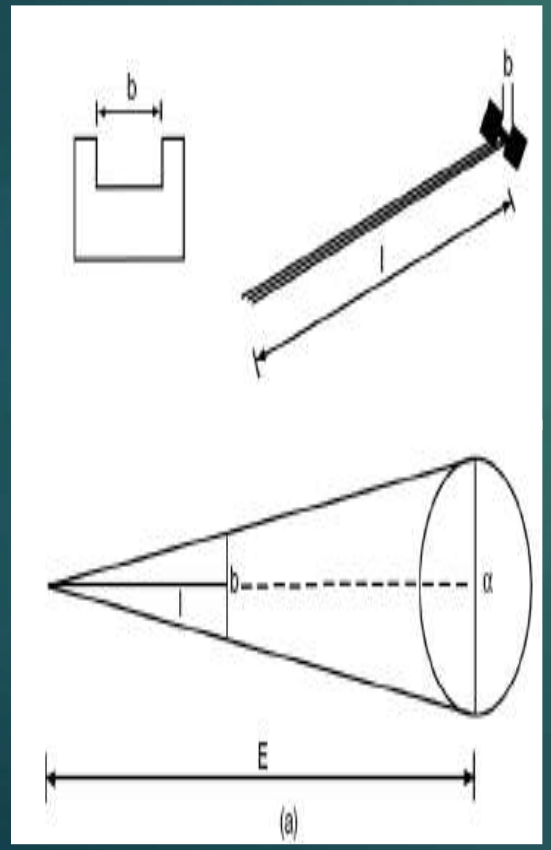
Tools: Relaskop, Prisms, Cruz-all, Bitterlich stick, Thumb, etc.

The probability of tallying a tree depends on its cross-sectional area and the sighting angle used.

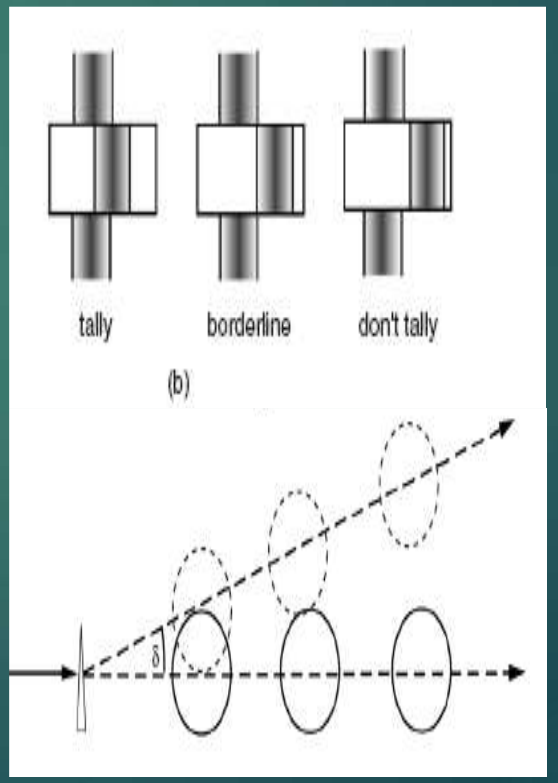
The smaller the sighting angle, the more trees included in the sample.

Types of instruments

Principle 1:
Divergent lines starting at the viewpoint and extending to a fixed reference distance and beyond until intercepted by an obstacle



Principle 2:
Diffraction of light rays, in our case diffraction of light rays from the tree as they go through a wedge prism in front of the observer



The Angle Gauge

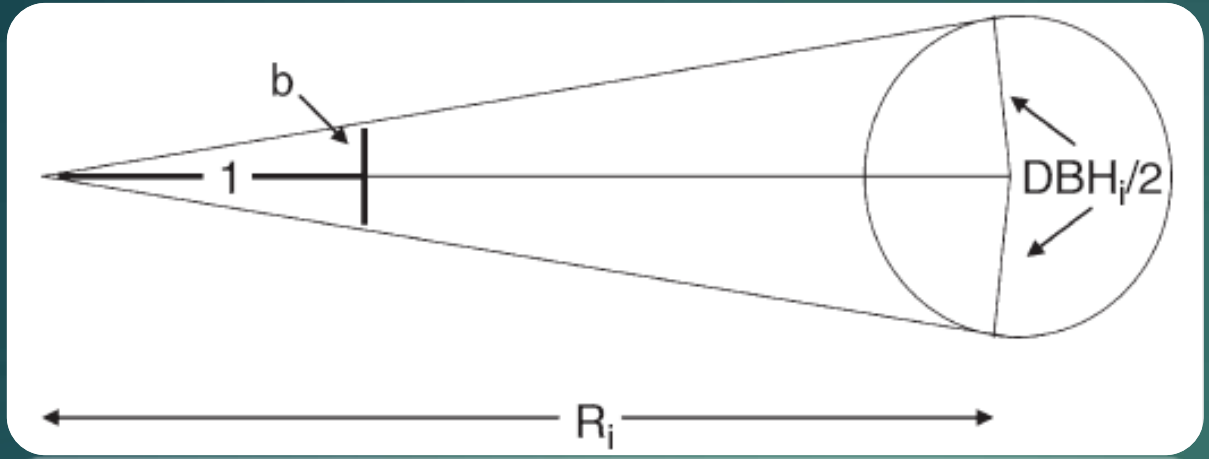








Geometric principle of Dendrometer



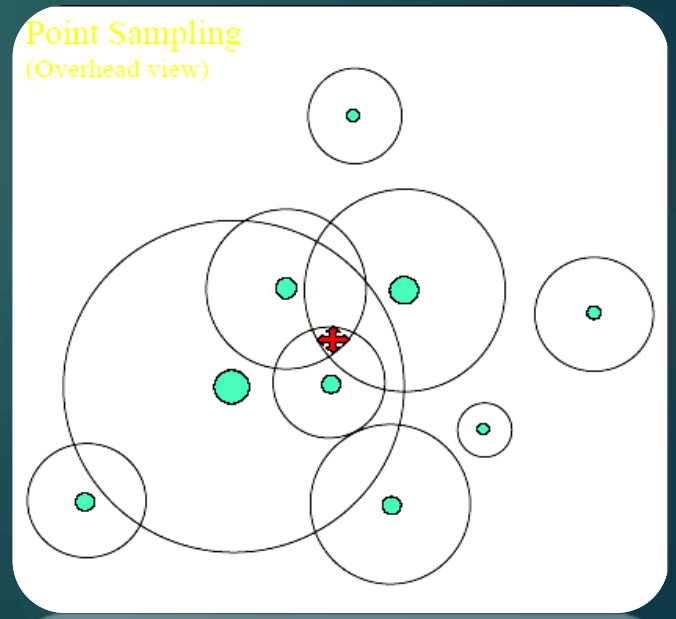
cross-arm 2-cm wide (b=2 cm) attached to a stick 1-m long (l=1 m) distance R_i with a DBH equal to DBH_i

$$\frac{b}{l} = \frac{2}{100} = \frac{\mathrm{DHB}_i}{R_i} \Leftrightarrow R_i = 50 \times \mathrm{DBH}_i$$

The sample area for ith tree

$$\pi \times (50 \times \mathrm{DBH}_i)^2$$

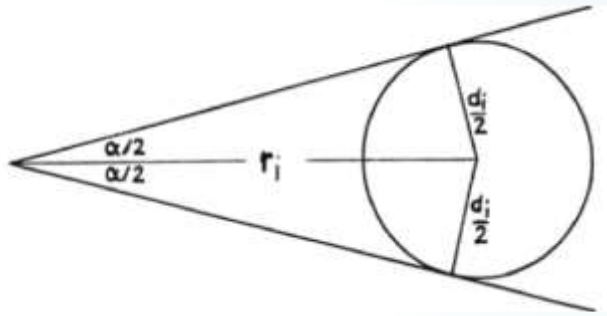
Each tree has a given plot size in relation to it's size.



$$\hat{G} = \frac{\pi/4 \times \mathrm{DBH}_i^{\,2}}{\pi \times \left(50 \times \mathrm{DBH}_i\right)^2} = \frac{1}{10000}\frac{\mathrm{m}^2}{\mathrm{m}^2} = 1\frac{\mathrm{m}^2}{\mathrm{ha}}$$

more general equation

$$\sin\frac{\alpha}{2} = \frac{d_i/2}{r_i} \quad \text{and} \quad \mathrm{r_i} = \frac{d_i}{2 \cdot \sin\frac{\alpha}{2}}$$



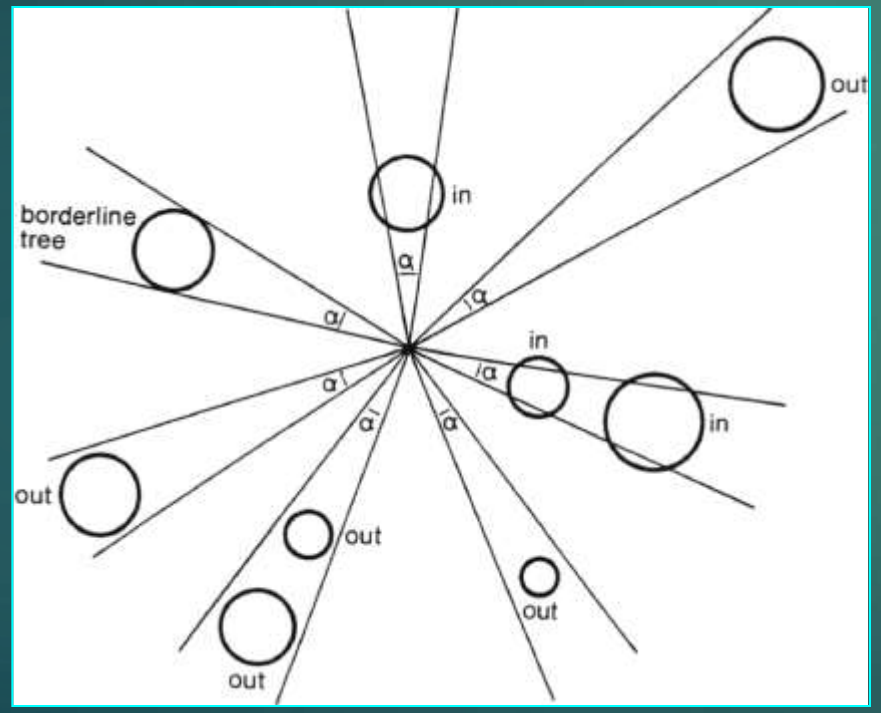
$$\mathrm{A} = \pi \cdot r^2 = \pi\frac{d_i^2}{4\left(\sin\frac{\alpha}{2}\right)^2}$$

$$\text{and } G_i\ (\mathrm{m^2/ha}) = 10^4 \cdot \frac{n_i \cdot \frac{\pi}{4} \cdot d_i^2}{\pi \cdot \frac{d_i^2}{4 \cdot \left(\sin\frac{\alpha}{2}\right)^2}} = 10^4 \cdot n_i \cdot \left(\sin\frac{\alpha}{2}\right)^2$$

$$\hat{G} = N_{\text{count}} \times 10^4 \times \sin^2(\alpha/2) = N_{\text{count}} \times \mathrm{BAF}$$

Point Sampling

Dendrometer (Overhead view)



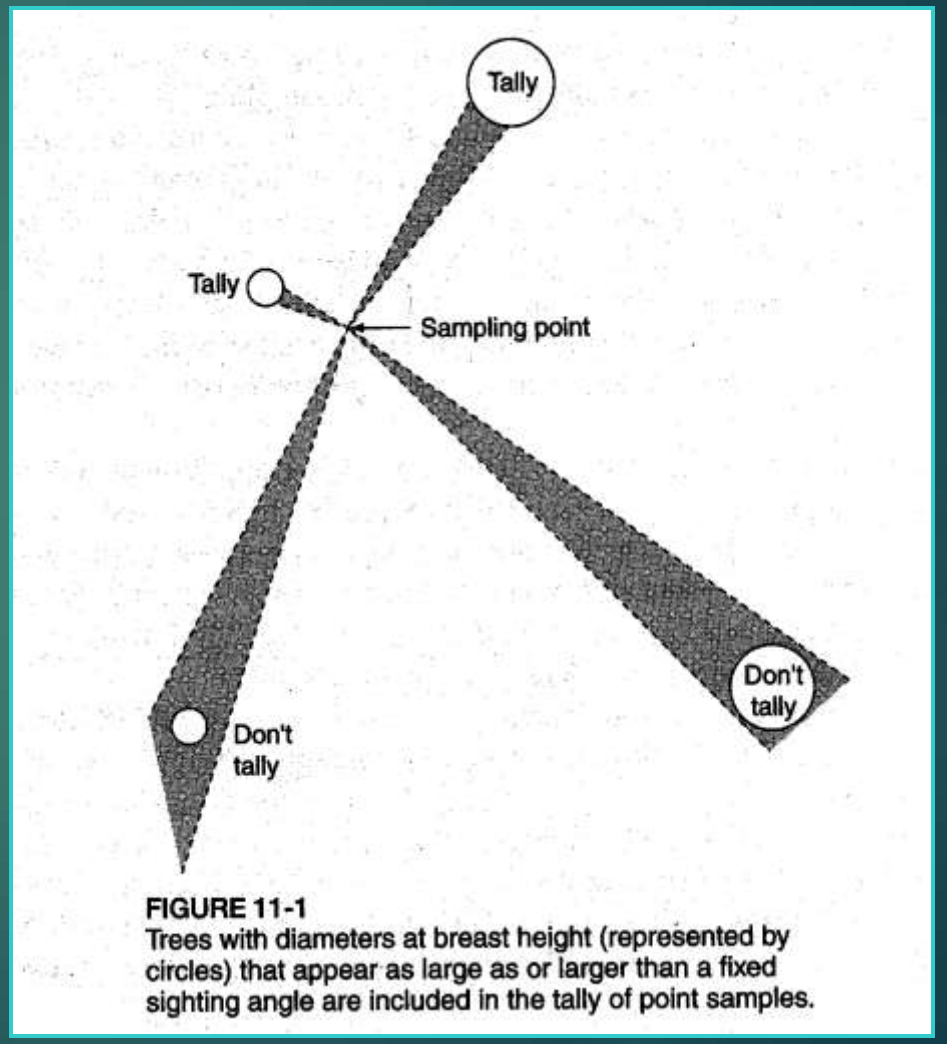
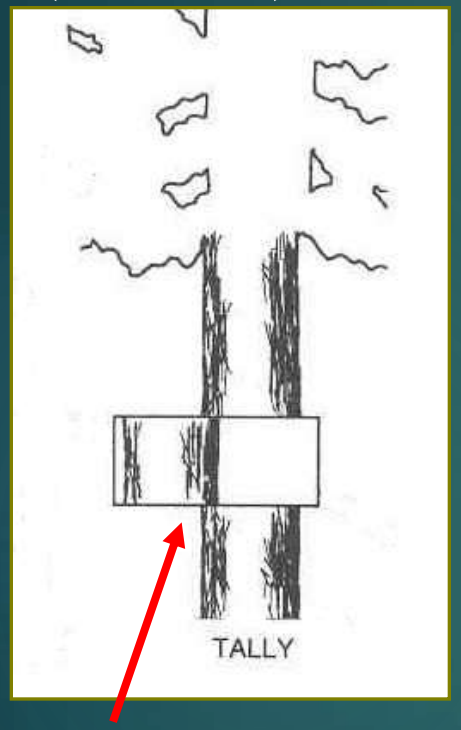


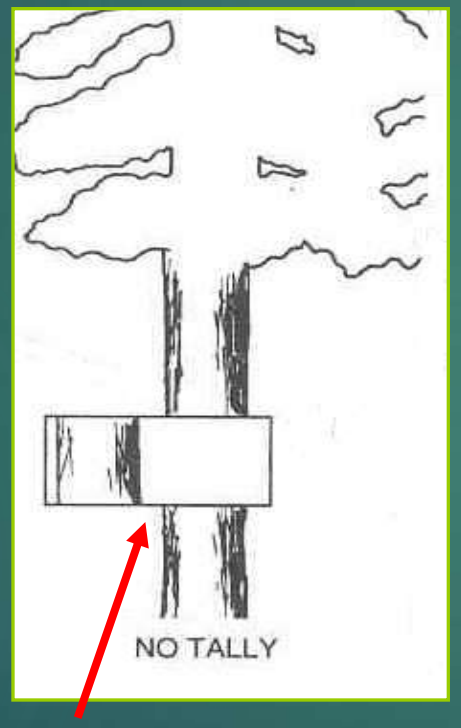
FIGURE 11-1
Trees with diameters at breast height (represented by circles) that appear as large as or larger than a fixed sighting angle are included in the tally of point samples.

Point Sampling

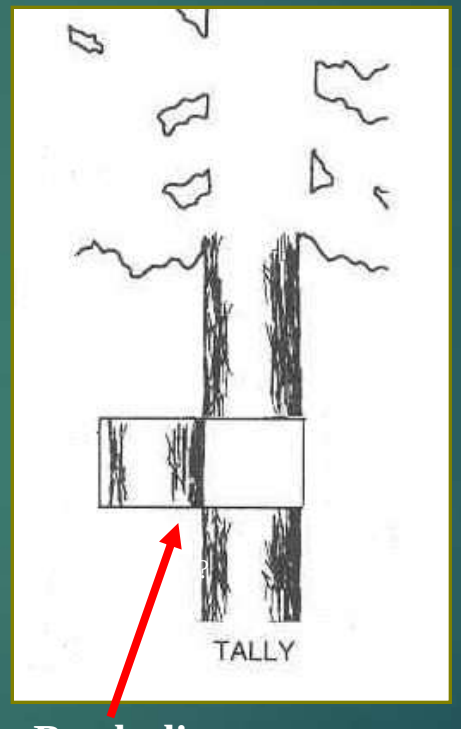
Using a prism (Side view)



Offset tree bole in prism does overlap actual tree bole



Offset tree bole in prism has no overlap with actual tree bole



Borderline tree - must measure distance from point center



Any attribute like Y

$$\hat{\bar{Y}}/\text{ha} = \frac{\text{BAF}}{n} \times \sum_{i=1}^{n} \sum_{j=1}^{N_{\text{count}}(i)} \frac{y_{ij}}{\text{BA}_{ij}}$$

stem count per hectare

$$\hat{N}/\text{ha} = \frac{\text{BAF}}{n} \times \sum_{i=1}^{n} \sum_{j=1}^{N_{\text{count}}(i)} \frac{1}{\text{BA}_{ij}}$$

basal area per hectare

$$\hat{\text{BA}}/\text{ha} = \frac{\text{BAF}}{n} \times \sum_{i=1}^{n} \sum_{j=1}^{N_{\text{count}}(i)} \frac{\text{BA}_{ij}}{\text{BA}_{ij}} = \frac{\text{BAF}}{n} \times \sum_{i=1}^{n} N_{\text{count}}(i)$$

Methods for Sampling Proportional to [Size]

All commonly used PPS techniques use sighting angles to select trees

Bitterlich sampling
Horizontal point sampling
Plotless sampling
Variable radius plot sampling (VRP)
Point sampling
Point sample plots
Poly-area plots
Prism plots, Prism cruising

Important Ideas about Point Sampling

1. **Basal Area Factor (BAF)**
   How much BA does each sampled tree represent?

2. **Plot Radius Factor**
   How far away can a tree be and still be sampled?

3. **Calculations from samples:**
   Basal area per acre
   Trees per acre
   Volume per acre

Basal area factor (BAF)

Related to the sighting angle size of the prism used in point sampling.

Chosen to provide a tally of 5 - 12 trees per point (8-10)

Eastern U.S.: 5, 10, 20

Western U.S.: 20, 40, 60

Plot radius factor (PRF)

The value to use in determining the maximum distance a tree of a given diameter can be from the point to still be tallied.

For example, when using a BAF 10 prism, the plot radius factor is 2.75.

Which means:
For each inch of dbh, a tree can be 2.75 feet from the point to still be included in the point's tally.

Determining the BAF of a prism, if unknown.....

Measure a tree, calculate its visible width (W) in feet.

Using a prism, back away from the tree until it becomes a “borderline” tree. Measure this distance (D) in feet.

Calculate the BAF of the prism:

$$BAF = 10{,}980\left(\frac{W}{D}\right)^2$$

Example: 12 inch tree (1 foot wide), moving back 33 feet from the tree we find it is a “borderline” tree in the prism...

$$BAF = 10{,}980\left(\frac{1}{33}\right)^2 = 10.08$$

Thumb as an angle gauge

Let thumb width = 0.85”
Eye to thumb distance = 25”



$$k = \frac{Diameter}{Plot_Radius} = \frac{0.85}{25.0} = 0.034 = gauge_constant$$

$$BAF = 10890 \cdot k^2 = 10890 \cdot (0.034)^2 = 12.59$$

BAF = 12.59 ft^2/acre

Calculating basal area from a point cruise:

$$BA = \left(\frac{Total\ trees\ tallied}{number\ of\ points} \right) BAF$$

Example:

- BAF 20 prism
- 402 trees tallied
- 25 points

$$BA = \left(\frac{402}{25} \right) 20 = 321.6\ \ sq.ft./acre$$

Two common approaches for Sampling

Basic approach:

Use 1 prism for entire cruise. Select BAF that tallies 7-13 trees per point. Measure DBH and height of all trees tallied. Use all trees tallied to calculate stand basal area, density, and volume.

Nested approach (Big BAF Method):

Use 1 prism with a BAF that tallies 7-13 trees per point as primary. At each sample point, use a second prism with a larger BAF that tallies 1-7 trees per point to select trees for diameter and height measurements. These trees are often called "volume trees" or "VBAR trees". Use primary tallies to calculate stand basal area. Use VBAR tree measurements to calculate stand density and volume.

Point-3P sampling

Method:

- At each point, sample trees are selected with a prism or angle gauge.
- Height of each tree is estimated.
- Each height estimate is paired with a random number from 3-P random number.
- Trees with a height estimated equal to, or larger than the random number are accurately measured for volume.
- For other tree only height estimation required.

$$\text{Tract volume} = \text{Area} * \text{BAF} * \overline{\Sigma H} * \left(\frac{\overline{\text{Vol}}}{\text{BA} * \text{H}}\right)$$

$$S^2_{\text{tract volume}} = \frac{S_1^2}{(NP) * (\overline{\Sigma H})^2} + \frac{S_2^2}{\frac{NP}{(\sum_{j=1} M_j)} * \left(\frac{\overline{\text{Vol}}}{\text{BA} * \text{H}}\right)^2}$$

where: Area = tract area

BAF = basal area factor of prism or angle gauge

$\overline{\Sigma H} = \sum_{j=1}^{NP} \sum_{i=1}^{Nj} H_{ij} / NP_i$ mean summed tree height per point

$\frac{\overline{\text{Vol}}}{\text{BA} * \text{H}} = \sum_{j=1}^{NP} \sum_{i=1}^{Mj} \left(\frac{Y_{ij}}{BA_{ij} * H_{ij}}\right) / M$; mean ratio of tree volume (or other tree characteristic) to product of basal area times height

NP = number of sampling points

N_j = number of trees selected with prism at point j

H_{ij} = estimated height of tree i at point j

M_j = number of trees selected with prism and 3-P criterion at point j

BA_{ij} = basal area of tree i at point j

Y_{ij} = volume (or other tree characteristic) of tree at point j

$M = \sum_{j=1}^{NP} M_j$; total number of trees selected with prism and 3-P criterion

S^2tract volume = approximate variance of tract volume as proportion of tract volume

S_1^2 = variance of summed tree height per point

S_2^2 = variance of ratio of tree volume (or other tree characteristic) to product of basal area times estimated height

منابع این جلسه

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